

Loan Gateway — Business Requirements Document

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Table of contents

1. Introduction & Scope
2. User Personas & Journeys
3. Sitemap & Page-by-Page Functional Specifications
4. Business Rules & Calculation Logic
5. API Contract
6. Data Model
7. Acceptance Criteria
8. Non-Functional Requirements
9. Out-of-Scope, Assumptions & Risks
10. Commercial Terms

Appendix A — Translation string sample Appendix B — Tracking-number format Appendix C — Glossary
Appendix D — Sample SQL DDL

1. Introduction & Scope

1.1 Background

[Customer] operates in the Bangladesh Non-Banking Financial Institution (NBFI) sector and wishes to deploy an online loan application gateway. A working prototype has been built externally and is deemed sufficient as the visual and behavioural baseline. This document articulates the requirements for [Vendor] to engineer a production-ready system aligned with the prototype.

1.2 Document purpose

This BRD serves three functions:

1. The authoritative scope statement for this engagement
2. The reference document for acceptance testing

3. The basis for fixed-price quotation

In any disagreement between the prototype and this BRD, **this BRD prevails.**

1.3 Canonical references

Reference	Location
Live prototype	https://rahilearns.github.io/Sample-Online-Platform/
Source code	https://github.com/Rahilearns/Sample-Online-Platform
Branding placeholder	The prototype uses "IDLC" throughout as a placeholder brand. Production branding to be supplied by [Customer].
Sample content	Every policy section in the prototype is marked "Sample content for prototype." Production policy text will be supplied by [Customer].

1.4 Goals & non-goals

Goals:

- Replicate the prototype's UX exactly in a production environment
- Establish backend infrastructure (database, admin, audit) supporting end-to-end loan application flow
- Comply with Bangladesh Bank's NBF digital service guidelines
- Bilingual (English + Bengali) from day one
- Mobile-first responsive design
- Audit trail for every submitted application

Non-goals (Phase 1):

- Real SMS gateway integration
- Real CIB/CIF lookup against [Customer]'s core banking system
- Document upload, OCR, e-signature
- Customer login portal / account self-service
- Push notifications, transactional email automation
- Mobile native apps (iOS / Android)

2. User Personas & Journeys

2.1 Primary personas

P1 — First-time loan applicant (mass-market) - Demographic: 25–45 years, salaried employee or small business owner, urban or semi-urban Bangladesh - Device mix: 60% mobile, 40% desktop - Language

preference: ~60% Bengali, ~40% English - Behaviour: scrolls quickly, abandons forms exceeding ~5 minutes, expects auto-fill, suspicious of being asked too much information up front - Primary pain points: existing NBFIs websites are slow, English-only, ask for too much information on the first screen, lack of feedback after submission

P2 — Existing IDLC customer - Demographic: previously borrowed from IDLC - Device mix: mostly mobile - Behaviour: expects the system to "know them" (auto-fill from CIF), values speed over discovery - Primary pain point: re-typing the same information that should already be on file

P3 — SME proprietor - Demographic: 30–55 years, partnership or proprietorship business owner - Device mix: desktop in office, mobile on the road - Behaviour: cares about working-capital cycle, decision-driven by interest rate + repayment terms; reviews documents carefully - Primary pain point: opaque pricing, slow approval cycles

P4 — Compliance / admin reviewer - Demographic: IDLC employee, branch officer - Behaviour: reviews submitted applications, processes documentation, follows up by phone with applicants - Primary pain point: scattered information across systems, manual data re-entry

2.2 User journeys

Journey 1 — First-time individual applicant (most common)

Step-by-step:

1. Visitor lands on home page (`index.html`) from a Facebook ad or Google search
2. Scrolls through the marketing content — hero, statistics, news, why-us
3. Clicks the persistent "Apply for Loan" button (bottom-right, sticky)
4. Lands on `apply.html` → picks "Loan for Individuals"
5. Lands on `apply-individual.html` → picks (e.g.) Home Loan
6. Modal opens — picks Purpose ("Apartment Purchase") + Financing Mode ("Conventional Financing")
7. Clicks Proceed → lands on `apply-home-loan.html?purpose=Apartment+Purchase&mode=Conventional`
8. Sees a two-column page: policy information (left) + application form + EMI calculator (right). The page header shows "Home Loan | Conventional Financing" and a chip below reading "Apartment Purchase".
9. Reads through the policy summary on the left (optional, builds trust)
10. Begins the application: types Name, NID, picks DOB via three dropdowns (Day / Month / Year), enters Phone
11. Clicks **Request OTP**. Four boxes appear in place of the button. Name/NID/DOB/Phone fields are locked.
12. Receives an SMS (Phase 2) — or for Phase 1, the screen shows "Demo OTP: 1234"
13. Types `1234` into the 4 boxes (auto-advance + paste handling)
14. Clicks **Submit OTP**. OTP card collapses, full application form reveals. Page scrolls to the top of the form-card.
15. Form auto-fills if NID matched an existing client; otherwise Name carries over and CIF/Email stay blank
16. Fills the remaining 5 required fields, answers "Loan Burden?" Yes/No, conditionally fills Monthly Loan Burden, enters Expected Loan Amount and Loan Term

17. Calculator on the right mirrors the entered amount + term at the standard 13% rate
18. Ticks both declaration checkboxes
19. Clicks **Submit Application**. Server-side eligibility check runs.
20. Receives one of three confirmations:
 - **Eligible at requested amount**: green panel, "Application Submitted", message references the requested amount, tracking number issued
 - **Reduced eligible amount (≥ BDT 25 lakh)**: green panel, "Application Submitted", message references the reduced amount with different wording, tracking number issued
 - **Ineligible**: amber panel, "Application Could Not Be Processed", standard rejection text, no tracking number
21. Visitor copies the tracking number for future reference (Phase 3: status lookup page)

Journey 2 — Existing customer

Identical to Journey 1 up to Step 11. After Step 14, the application form auto-fills Name, CIF, and Email from the customer lookup, eliminating ~30 seconds of friction. The visitor only types the new application-specific fields (income, burden, expected amount, loan term).

Journey 3 — SME proprietor

Same shape as Journey 1 but starts from `apply-business.html`, with the modal asking for Desired Product (Purnota / Working Capital / Long Term Loan) instead of Purpose. Lands on `apply-sme-loan.html`. Content emphasis on business documents.

Journey 4 — Corporate applicant (out-of-prototype)

Visitor clicks "Loan for Corporates" → browser opens `https://idlc.com/corporate-loan` in a new tab. Visitor is routed outside this gateway entirely. No internal Corporate page exists or is needed.

Journey 5 — Admin reviewer

Visitor (IDLC employee) logs into the admin dashboard (Phase 1 includes login). Lands on the application list — paginated, sortable by date / status / amount. Searches by tracking number, opens row, reviews all fields. Exports the day's applications as CSV. Phase 3 will add status workflows, document review, and customer communication tools.

3. Sitemap & Page-by-Page Functional Specifications

3.1 Sitemap

```
index.html (marketing home)
├─ floating "Apply for Loan"
│   └─ apply.html
│       ├── apply-individual.html
│       │   └─ modal: Purpose × Mode →
│       │       ├── apply-home-loan.html
│       │       ├── apply-car-loan.html
│       │       └─ apply-personal-loan.html
│       └─ apply-business.html
│           ├── modal: Product × Mode → apply-sme-loan.html
│           └─ external → idlc.com/corporate-loan (new tab)

/admin (Phase 1)
├─ /admin/login
└─ /admin/applications
    ├── /admin/applications/:tracking_number
    └─ /admin/applications/export.csv
```

3.2 Page-by-page specifications

index.html — Marketing home

- Persistent disclaimer ribbon at top (production: replaced or removed)
- Top navigation: brand mark (square red "I" + "IDLC"), 7 menu items (About Us, Why Choose Us, News, Career, Investor Relations, Sustainability), search icon, Login link
- Hero section: large headline + sub + two CTA buttons (decorative on home; functional CTAs are floating)
- Stat strip with 4 KPIs in a red band
- News & updates strip — 4 cards
- "Why choose us?" band on dark green/red
- Quick links — 4 cards (Monthly Business Review, Careers, Customer Portal, Stock Ticker)
- Footer with 4 link columns + city skyline illustration + social icons
- Floating bottom-right: **Customer Login** (outlined) + **Apply for Loan** (filled) — sticky, must remain visible while scrolling. Clicks inside `.home-content` are inert via `pointer-events: none`. Only the floating buttons and language toggle are clickable.
- Language toggle (EN / বাংলা) fixed top-right; selection persists via localStorage

apply.html — Individual / Business chooser

- Header: brand + back link to home + language toggle

- Centered eyebrow chip ("APPLY FOR A LOAN"), centered title ("What kind of loan are you looking for?"), centered subtitle
- Two large cards (Loan for Individuals, Loan for Businesses) with icons, descriptions, no Continue button (entire card is clickable)
- Footer note: "Your information is encrypted and reviewed only by authorised officers. **BB-regulated.**"

`apply-individual.html` — Home / Car / Personal chooser

- Three cards (Home Loan, Car Loan, Personal Loan)
- Each card has a small line-icon and the product name
- Click triggers a modal — see §3.3 for modal spec
- Modal Proceed navigates to the corresponding product page with `?purpose=...&mode=...` URL params

`apply-business.html` — SME / Corporate chooser

- Two cards (Loan for SME, Loan for Corporates)
- SME card click triggers a modal (same shape as Individual modal but with different dropdown content — see §3.3)
- Corporate card is a regular `<a href="https://idlc.com/corporate-loan" target="_blank" rel="noopener noreferrer">`. No modal, no internal navigation, no inline JS interception.

Product pages — `apply-{home,car,personal,sme}-loan.html`

Two-column layout (1600px max-width on desktop, stacks on mobile under 1200px):

Left column (`product-info-col` , fixed 600px on desktop): Six policy sections in order: 1. Purpose — what the loan is for 2. Eligibility — applicant criteria 3. Policy Highlights — loan amount, tenor, rate, fees 4. Required Documents — checklist 5. Terms & Conditions — numbered list 6. Frequently Asked Questions — accordion (HTML `<details>` elements)

Each section ends with the disclaimer note (production: removed/replaced).

Right column (`product-action-col` , ~870px on 1600px): Two stacked cards: 1. Application Form — multi-stage (see §3.4) 2. EMI Calculator — three text inputs (Amount, Rate, Tenor) + live results panel showing Monthly EMI, Total Interest, Total Payment. No upper cap on any input. See §3.5.

Header: Eyebrow shows product name (e.g. "HOME LOAN"). H1 reads "Home Loan | Conventional Financing" — the financing-mode suffix is appended from URL params. Below H1, a red chip shows the chosen purpose (e.g. "Apartment Purchase").

Admin pages (Phase 1)

- `/admin/login` — email + password form, server-side auth, session cookie
- `/admin/applications` — paginated list (20 per page), columns: tracking number, applicant name, product, applied amount, eligible amount, status, submitted at. Filter by status. Search by tracking number, NID, or phone. Export current view as CSV.

- `/admin/applications/:tracking_number` — full application detail, audit log

3.3 Modal specification

The modal is identical in structure across `apply-individual.html` and `apply-business.html` :

- Dimmed full-screen backdrop (closes on backdrop click)
- Centered dialog max-width 460px, padding 32px, rounded 18px
- Close button (×) top-right
- Pencil-sketch illustration at the top (product-specific SVG)
- Title — "Tell us about your [Product] Loan"
- Subtitle — "Help us route your application to the right team."
- Two dropdowns:
- **Individual variant:** Purpose of the Loan × Financing Mode. Purpose options come from a static data structure (see Appendix A or `translations.js`). 6 options for Home, 1 for Car, 8 for Personal.
- **Business variant:** Choose Your Desired Product × Financing Mode. Product options: Purnota / Working Capital / Long Term Loan.
- Both dropdowns start blank — no auto-selected first option, no placeholder text shown
- Financing Mode is always 2 options: Conventional Financing / Islamic Financing
- Proceed button at the bottom of the dialog — full-width red pill button
- Proceed validates both dropdowns have a value before navigating
- Esc key closes the modal

3.4 Multi-stage application form specification

Stage 1 — Initial form (OTP gate)

Fields, all required, all marked with red asterisk in label: - Name (text) - NID (text, pattern 10/13/17 digits, JavaScript-validated) - Date of Birth — three dropdowns (Day, Month, Year): - All three start blank - Year populated by JavaScript: most recent first (currentYear – 18) down to (currentYear – 100) - Month is a static 12-option dropdown - Day is populated by JavaScript with 1–31 by default, narrows to the month's max when Month is picked, narrows further to honour leap years when Year is picked. Day auto-clamps if a previously selected value exceeds the new max. - Phone (tel, pattern `01[0-9]{9}` , 11 digits)

Submit button: "Request OTP"

Validation runs on submit. If any field is invalid, focus moves to the first invalid field and a red toast appears with a field-specific message. If all valid, the OTP card slides into view.

Stage 2 — OTP card

- 4 single-digit boxes, monospace font, auto-advance, paste-the-whole-OTP support, backspace navigation to previous box
- Demo OTP hint: "Demo OTP: `1234` " (Phase 1 only; remove in Phase 2)
- Submit button morphs from "Request OTP" to "Submit OTP"

- On submit: if all 4 digits = 1234 → Stage 3; else clear boxes, show "Incorrect OTP. Please try again." in red, refocus first box

Stage 3 — Full application form

Revealed in place of Stage 1+2. Auto-scrolls to top of form-card.

Fields:

1. **Name** (text, required) — pre-filled from OTP gate, or with hardcoded existing-client name if NID matched
2. **Customer No. (CIF)** (text, optional) — pre-filled if existing client, otherwise blank
3. **Email** (email, required) — pre-filled if existing client, otherwise blank
4. **Profession** (dropdown, required) — options: Salaried / Business Owner / Self-Employed / Other
5. **Total Monthly Income (BDT)** (text, required, auto-comma formatted as user types) — spans full width
6. **Do you have any loan burden currently?** (radio, required, Yes/No)
7. **Monthly Loan Burden (BDT)** (text, required if previous answered Yes, auto-comma) — conditionally revealed
8. **Expected Loan Amount from IDLC (BDT)** (text, required, auto-comma)
9. **Loan Term (years)** (text, required)

Below the fields: two declaration checkboxes, both required:

1. *"I declare that the information provided above is true and provided in my sound mind."*
2. *"I hereby authorize IDLC to obtain my credit information from Credit Information Bureau of Bangladesh Bank and authorize IDLC to use all the information in processing the applied loan."*

At the very top of Stage 3: a horizontal progress bar showing % of required fields completed (CIF excluded from counted required when conditional checkboxes resolve appropriately).

Submit button: "Submit Application". Disabled until both declarations are ticked? No — the button is always enabled, but clicking it without ticked declarations shows a red toast and focuses the first unticked checkbox.

3.5 EMI Calculator specification

Located in the right column of every product page, below the application form.

Inputs (3 free-form text fields, no upper cap):

- Loan Amount (BDT) — auto-comma formatted as user types
- Interest Rate (%) — accepts decimals (e.g. 13.5)
- Tenor (years) — integer

Outputs (read-only, update live within 100ms of any input change):

- Monthly EMI
- Total Interest

- Total Payment

Formula: $EMI = P \times r \times (1+r)^n / ((1+r)^n - 1)$ where P = principal (BDT), r = monthly rate (annual / 12 / 100), n = tenor in months (years × 12).

Form-to-calc sync: When the visitor types into the application form's "Expected Loan Amount" or "Loan Term" fields, those values mirror into the calculator's Amount and Tenor inputs, and the calculator's Rate auto-sets to 13%. After this auto-sync the visitor can override any calculator input without affecting the form.

Initial state: All inputs blank with 0 placeholder. Outputs show ₹ 0.

4. Business Rules & Calculation Logic

4.1 Eligibility decision matrix

Parameter	Value
Standard interest rate (eligibility math)	13.00% per annum
Threshold	Total monthly load ≤ 70% of declared monthly income
Reduced-amount minimum	BDT 25,00,000 (25 lakh)
Currency	BDT throughout

Algorithm (pseudocode):

```

income          = numeric value of "Total Monthly Income"
hasBurden        = radio "Do you have any loan burden currently?" == "yes"
monthlyBurden    = numeric value of "Monthly Loan Burden" if hasBurden else 0
expectedAmount   = numeric value of "Expected Loan Amount from IDLC"
years            = numeric value of "Loan Term (years)"
maxLoad          = income × 0.70

# Existing burden alone overshoots the threshold
if monthlyBurden > maxLoad:
    return { status: "ineligible" }

newEmi = computeEMI(expectedAmount, 13.00, years)
totalLoad = newEmi + monthlyBurden

if totalLoad <= maxLoad:
    return { status: "eligible", amount: expectedAmount }

# Solve for the largest principal that fits the remaining room
roomForNewEmi = maxLoad - monthlyBurden
maxLoan = computeMaxLoan(roomForNewEmi, 13.00, years)
maxLoan = floor(maxLoan / 1000) * 1000 # round down to nearest BDT 1,000

if maxLoan >= 2_500_000:
    return { status: "reduced", amount: maxLoan }

return { status: "ineligible" }

```

Inverse formula (for computing max principal from a target EMI ceiling):

$$P_{\text{max}} = \text{EMI}_{\text{target}} \times ((1+r)^n - 1) / (r \times (1+r)^n)$$

4.2 Confirmation message templates

Eligible (requested fits):

"Congratulations, you are eligible for the requested loan amount of BDT **[amount]**, subject to authenticity of your given information, further credit assessment from IDLC, and authenticity & validity of the required documents."

Reduced (eligible < requested, but ≥ BDT 25 lakh):

"Congratulations, you are eligible for a loan amount of BDT **[amount]**, subject to authenticity of your given information, further credit assessment from IDLC, and authenticity & validity of the required documents."

Ineligible:

"Your monthly income will not cover your loan burden and IDLC's risk appetite. We are unable to process this application at the requested amount."

Both eligible and reduced statuses receive a tracking number. Ineligible receives no tracking number.

Both templates exist in English and Bengali — see `translations.js` (keys `eligibility_eligible_html`, `eligibility_reduced_html`, `eligibility_ineligible`, `app_submitted_title`, `app_ineligible_title`).

4.3 OTP rules

Rule	Phase 1	Phase 2
OTP value	Hardcoded 1234	Random 4-digit, dispatched via SMS gateway
Expiry	n/a (always valid)	5 minutes
Max retries	n/a	3 per phone per hour
Send rate-limit	n/a	5 per phone per hour, 3 per IP per minute
Delivery channel	Demo hint shown on UI	Bangladesh-licensed SMS gateway (vendor recommendation; options: SSLWireless, SMS Express, Robi Axiata corporate SMS)

4.4 Existing-client lookup

Rule	Phase 1	Phase 2
Trigger	OTP success	OTP success
Source	Hardcoded mapping	[Customer]'s CIF / core banking API
Match key	NID	NID + DOB (composite for security)
Fields returned on match	Name, CIF, Email	Name, CIF, Email (+ extensible)
Behaviour on match	Auto-fill silently in the application form	Same
Behaviour on no match	Carry over typed Name from OTP gate; CIF + Email blank	Same
UX side-effect	None (no notification, no badge)	Same

4.5 Validation rules

Field	Pattern / Rule
Name	Non-empty, trimmed
NID	<code>^[0-9]{10,17}\$</code>
DOB	Valid Gregorian date; applicant $\geq$ 18 years on date of OTP request; $\leq$ 100 years old
Phone	<code>^01[0-9]{9}\$</code> (Bangladesh mobile, 11 digits)
OTP	4 digits, all from 0-9
Email	Standard HTML5 email pattern
Profession	One of: salaried, business, self, other
All monetary fields	Non-negative integer in BDT
Loan Term	Positive integer (years)
Declarations	Both must be checked

4.6 Tracking number generation

- Format: `IDLC-YYMMDD-XXXX`
- `IDLC` — fixed brand prefix (production: replaced with [Customer]'s production brand)
- `YYMMDD` — date of application (2-digit year, month, day)
- `XXXX` — 4-char alphanumeric, uppercase, excluding `0`, `0`, `1`, `I`, `L` to avoid visual ambiguity
- Generated server-side, atomically, with database UNIQUE constraint on the `tracking_number` column
- Generated only on `eligible` and `reduced` statuses
- Example: `IDLC-260520-K7P4`

5. API Contract

All endpoints: - Return JSON - Use HTTPS (TLS 1.2+) - Use bearer-token authentication post-OTP (token issued by `/api/otp/verify`) - Validation errors → HTTP 400 with `{errors: [{field, message}]}` - Server errors → HTTP 5xx with `{error_id}` (for audit log lookup)

Endpoint	Method	Auth	Purpose
/api/otp/request	POST	none	Initiate OTP for a phone+name+NID+DOB combination
/api/otp/verify	POST	none	Verify entered OTP; issue session bearer token
/api/customer/lookup	GET	bearer	NID-based existing-client check
/api/applications	POST	bearer	Submit full application; run eligibility; return result
/api/applications/:tracking	GET	bearer or admin	Read application by tracking number
/api/admin/applications	GET	admin	Paginated list with filters
/api/admin/applications/export	GET	admin	CSV export

5.1 POST /api/otp/request

Request:

```
{
  "name": "Md. Rakib Hasan",
  "nid": "5102284394",
  "dob": "1990-05-15",
  "phone": "01712345678"
}
```

Response (Phase 1 — always success):

```
{ "success": true, "expires_in": 300 }
```

5.2 POST /api/otp/verify

Request:

```
{ "phone": "01712345678", "otp": "1234" }
```

Response (success):

```
{ "success": true, "token": "<JWT>" }
```

Response (failure):

```
{ "success": false, "error": "incorrect_otp" }
```

5.3 GET /api/customer/lookup

Request: ?nid=5102284394 (Authorization: Bearer)

Response (match):

```
{
  "exists": true,
  "name": "Mahfuzul Islam",
  "cif": "553577",
  "email": "Imahfuzul@gmail.com"
}
```

Response (no match):

```
{ "exists": false }
```

5.4 POST /api/applications

Request (full payload):

```
{
  "product": "home-loan",
  "purpose": "Apartment Purchase",
  "financing_mode": "Conventional",
  "name": "Md. Rakib Hasan",
  "nid": "5102284394",
  "dob": "1990-05-15",
  "phone": "01712345678",
  "cif": "553577",
  "email": "Imahfuzul@gmail.com",
  "profession": "salaried",
  "monthly_income": 200000,
  "has_burden": false,
  "monthly_burden": null,
  "expected_amount": 2500000,
  "loan_term_years": 5,
  "declaration_1": true,
  "declaration_2": true,
  "language": "en"
}
```

Response (eligible):

```
{
  "tracking_number": "IDLC-260520-K7P4",
  "status": "eligible",
  "eligible_amount": 2500000
}
```

Response (reduced):

```
{
  "tracking_number": "IDLC-260520-Q2RN",
  "status": "reduced",
  "eligible_amount": 3070000
}
```

Response (ineligible):

```
{
  "tracking_number": null,
  "status": "ineligible",
  "eligible_amount": 0
}
```

5.5 GET /api/admin/applications

Query parameters: `?page=1&size=20&status=eligible&product=home-loan&q=rakib`

Response:

```
{
  "total": 142,
  "page": 1,
  "size": 20,
  "applications": [ { ... }, { ... } ]
}
```

6. Data Model

6.1 Tables

applications — primary record

Column	Type	Notes
id	UUID PRIMARY KEY	
tracking_number	VARCHAR(20) UNIQUE	NULL if status = ineligible
product	ENUM	home-loan, car-loan, personal-loan, sme-loan
purpose	VARCHAR(120)	
financing_mode	ENUM	Conventional, Islamic
name	VARCHAR(120) NOT NULL	
nid	VARCHAR(20) NOT NULL	INDEX
dob	DATE NOT NULL	
phone	VARCHAR(15) NOT NULL	INDEX
cif	VARCHAR(20) NULLABLE	
email	VARCHAR(120)	
profession	ENUM	salaried, business, self, other
monthly_income	DECIMAL(15,2)	
has_burden	BOOLEAN	
monthly_burden	DECIMAL(15,2) NULLABLE	
expected_amount	DECIMAL(15,2)	
loan_term_years	INT	
eligibility_status	ENUM	eligible, reduced, ineligible
eligible_amount	DECIMAL(15,2)	0 if ineligible
language	ENUM	en, bn
declarations_accepted_at	TIMESTAMP	
ip_address	VARCHAR(45)	IPv4 or IPv6
user_agent	TEXT	
created_at	TIMESTAMP DEFAULT NOW()	
updated_at	TIMESTAMP DEFAULT NOW() ON UPDATE NOW()	

audit_log — append-only

Column	Type
id	UUID PRIMARY KEY
application_id	UUID FK → applications.id
event_type	VARCHAR(40)
actor	VARCHAR(120) — "system", admin email, or NULL
timestamp	TIMESTAMP
payload	JSON

otp_requests — short-lived, can be purged after 24 hours

Column	Type
id	UUID PRIMARY KEY
phone	VARCHAR(15)
code_hash	VARCHAR(64) — never store the raw OTP
requested_at	TIMESTAMP
expires_at	TIMESTAMP
verified_at	TIMESTAMP NULLABLE
attempt_count	INT DEFAULT 0

admin_users — Phase 1 admin login

Column	Type
id	UUID PRIMARY KEY
email	VARCHAR(120) UNIQUE
password_hash	VARCHAR(120) — bcrypt or argon2
role	ENUM (admin, reviewer)
created_at	TIMESTAMP
last_login_at	TIMESTAMP NULLABLE

6.2 Relationships

- applications.id → audit_log.application_id (1-to-many)
- applications.phone → otp_requests.phone (loose link, not FK)

See Appendix D for sample DDL.

7. Acceptance Criteria

Each AC is demonstrable in under 60 seconds on the staging environment.

#	Criterion
AC-1	Open / → disclaimer ribbon visible at top; brand, lang toggle, hero, stat strip, news cards, why-us band, quick links, footer all render; floating Customer Login + Apply for Loan pinned bottom-right; scrolling does not move them.
AC-2	Click বাংলা on any page → all visible UI translates → click any internal link → still in Bengali → refresh → still in Bengali.
AC-3	From /apply-individual.html click Home Loan → modal opens with house sketch, blank Purpose dropdown (6 options), blank Financing Mode dropdown (2 options). Pick "Apartment Purchase" + "Conventional Financing" → click Proceed → land on apply-home-loan.html?purpose=Apartment+Purchase&mode=Conventional . Title reads "Home Loan Conventional Financing". Chip below reads "Apartment Purchase".
AC-4	From apply-business.html click Loan for SME → modal opens with storefront sketch, dropdowns Purnota / Working Capital / Long Term Loan × Conventional / Islamic Financing.
AC-5	From apply-business.html click Loan for Corporates → new browser tab opens https://idlc.com/corporate-loan .
AC-6	On any product page, enter NID 5102284394 , DOB valid date ≥18 yrs, valid BD phone → Request OTP → 4 OTP boxes appear → enter 1234 → Submit OTP → loan form reveals with Name="Mahfuzul Islam", CIF="553577", Email="lmahfuzul@gmail.com" pre-populated.
AC-7	Same flow with any NID other than 5102284394 → after OTP, loan form reveals with Name = whatever was typed in OTP gate, CIF and Email blank.
AC-8	Pick DOB making applicant <18 → click Request OTP → red toast: "Applicants must be at least 18 years of age to submit this loan application." OTP does not fire.
AC-9	DOB Day=31, Month=February, Year=2000 → Day auto-clamps to 29 (leap year). Change Year to 2001 → Day clamps to 28.
AC-10	After OTP, enter Income=200,000; Has Burden=No; Expected Amount=20,00,000; Loan Term=5 → tick both declarations → Submit. Result: green success, message reads "... requested loan amount of BDT 20,00,000...", tracking number visible.
AC-11	After OTP, enter Income=1,00,000; Has Burden=No; Expected Amount=5,00,00,000; Loan Term=5 → Submit. Result: green success, message reads "... a loan amount of BDT [computed reduced]...", tracking number visible.
AC-12	After OTP, enter Income=30,000; Has Burden=Yes; Monthly Loan Burden=25,000; Expected Amount=25,00,000; Loan Term=5 → Submit. Result: amber warning, message reads "Your monthly income will not cover your loan burden and IDLC's risk appetite...", no tracking number.
AC-13	Try to Submit Application without ticking both declarations → red toast: "Please accept both declarations to continue." Focus moves to first unticked checkbox.
AC-14	Type 2500000 into any BDT field → formats live to 25,00,000 . Mid-string editing preserves cursor position.

#	Criterion
AC-15	Type <code>100000000</code> (10 crore) into EMI calculator Loan Amount → result rows update within 100ms; no upper cap.
AC-16	Admin login → see paginated applications list, search by tracking number, view row detail. Export current page as CSV with all columns.
AC-17	Submit any eligible application → confirmation rendered in current language → toggle to বাংলা → all confirmation text (including tracking-number block) re-renders in Bengali.
AC-18	Submit an eligible application → check the audit log table for: <code>application_created</code> , <code>eligibility_checked</code> , <code>tracking_number_issued</code> rows.
AC-19	Attempt OTP verification with code other than <code>1234</code> → server returns 400 with <code>incorrect_otp</code> . OTP card shows error message; boxes clear; first box re-focused.
AC-20	Page-by-page Lighthouse mobile audit: performance ≥ 85, accessibility ≥ 90, best practices ≥ 90, SEO ≥ 90.

8. Non-Functional Requirements

8.1 Performance

- Time to First Contentful Paint (FCP) on 3G mobile: **< 2.5s**
- Total Blocking Time (TBT): **< 300ms**
- Form submission round-trip on Phase 1 stub: **< 800ms**
- 95th percentile response time for any API: **< 500ms**
- Page weight per route: **< 500 KB transferred** (excluding fonts)

8.2 Security

- HTTPS everywhere — Let's Encrypt acceptable, must auto-renew
- OWASP Top 10 mitigations
- Server-side validation on every input (do not trust client-side validation)
- NID treated as sensitive PII — encrypted at rest, never written to access logs
- Rate limiting: max 5 OTP requests per phone per hour, 3 per IP per minute
- Session timeout: 30 minutes idle
- CSRF tokens on all POST endpoints
- Strict Content Security Policy headers
- HSTS headers with `max-age=31536000; includeSubDomains; preload`
- All database queries parameterized (no string concatenation)
- Passwords hashed with bcrypt (cost 12+) or argon2id

- No PII in URL paths, query strings, log statements, or analytics events

8.3 Compliance

- Bangladesh Bank Digital Banking and NBFI service guidelines (vendor to confirm latest version applicable)
- Data residency: customer data must reside on servers physically located in Bangladesh
- Audit trail: every submitted application logged with timestamp, IP, user-agent, full payload at time of submission
- Retention: applications retained for 7 years per BB requirements
- KYC compliance hooks (Phase 2 — vendor must architect the schema to accommodate KYC fields even in Phase 1)

8.4 Accessibility

- WCAG 2.1 Level AA compliance
- Keyboard navigation for all flows (Tab, Shift+Tab, Enter, Esc)
- Screen-reader compatibility (aria-* attributes already present in the prototype — preserve)
- Colour contrast ratio $\geq 4.5:1$ for body text, $\geq 3:1$ for UI elements
- Focus indicators on every interactive element
- Form errors announced to assistive technology

8.5 Browser support

- Chrome 90+ (desktop, Android)
- Safari 14+ (desktop, iOS)
- Firefox 90+
- Edge 90+
- Internet Explorer **not required**

8.6 Hosting & SLA

Metric	Phase 1 target	Phase 2 target
Uptime	99.5% (single instance)	99.9% (with redundancy + standby)
Daily DB backup	Required, 7-day retention	Required, 30-day retention
Weekly full backup	Required, 90-day retention	Required, 365-day retention
Monitoring	Basic uptime + error-rate	Full APM (response times, error rates, throughput)
Alert channels	Email + SMS to [Customer] contact	+ on-call rotation

8.7 Internationalisation

- English + Bengali only in Phase 1
 - All UI strings must be sourced from a single translation file (`translations.js` in the prototype). No hardcoded English in HTML or rendered server output.
 - Date formatting respects the active language
 - Number formatting (commas) uses Bangladeshi style (12,34,567) in both languages
 - Bengali font: Hind Siliguri preferred (already in the prototype)
-

9. Out-of-Scope, Assumptions & Risks

9.1 Out of scope (Phase 1)

- Real SMS gateway integration
- Real CIF / NID lookup against [Customer]'s core banking system
- Document upload, OCR, e-signature, e-KYC
- Push notifications
- Transactional email automation (welcome emails, status updates)
- Customer Login flow / account self-service / loan-status portal for customers
- Mobile native apps (iOS, Android)
- Multi-tenancy (one institution = one deployment)
- Marketing analytics integration (GA4, Facebook Pixel) — vendor adds Phase 3 if requested
- A/B testing infrastructure

9.2 Assumptions

ID	Assumption
A1	[Customer] will provide a production domain + DNS access
A2	[Customer] will provide brand assets (logo, final colour palette, brand guidelines) before Phase 1 acceptance demo
A3	[Customer] will provide the production SSL certificate or approve Let's Encrypt
A4	All production content (policy text, T&Cs, FAQs, real address, real phone, real stock-ticker values, news items) will be supplied by [Customer] in editable format before launch
A5	SMS gateway selection and contract is [Customer]'s responsibility
A6	CIB/CIF API endpoint specifications will be provided by [Customer] before Phase 2 kickoff
A7	[Customer] designates a single project owner empowered to make scope and acceptance decisions
A8	Acceptance testing performed by [Customer]; vendor provides UAT support but is not responsible for [Customer]'s internal test cycle duration
A9	The placeholder content in the prototype (sample policy, sample T&Cs, sample address, sample stock prices, sample news) carries no IP rights or legal weight; [Customer] will replace all of it before launch
A10	The "IDLC" branding in the prototype is a working placeholder. Production deployment will use [Customer]'s actual brand. Any external link in the prototype (e.g. idlc.com/corporate-loan) will be reviewed by [Customer] for production.

9.3 Risks

ID	Risk	Mitigation
R1	SMS gateway integration takes longer than estimated	Phase 2 is quoted separately; Phase 1 uses stubs
R2	CIB/CIF API spec from [Customer] is incomplete	Phase 1 schema accommodates lookup; vendor's Phase 2 quote can be revised
R3	Bangladesh Bank guidelines change mid-engagement	Compliance hooks designed for extensibility; vendor accommodates within reason
R4	Browser support edge cases (older Android browsers)	Vendor targets evergreen browsers; degradation handled gracefully
R5	Performance on rural 2G/3G	Static front-end + minified assets + small page weight target (\$8.1)
R6	Scope creep mid-Phase-1	Change-request process: any change requires written approval + new fixed-price addendum

10. Commercial Terms

10.1 Phasing & quotation

Vendor is asked to quote **Phase 1 (fixed price)** at this stage. Phase 2 and Phase 3 are scoped at a high level here but will be re-scoped and quoted separately at the appropriate time.

10.2 Payment milestones (Phase 1)

Milestone	Trigger	%
M1 — Kickoff	Signed contract + technical design document (≤ 10 pages) submitted by vendor	30%
M2 — Phase 1 acceptance	All 20 ACs demonstrated on staging environment + [Customer] sign-off	40%
M3 — Warranty close	30 days post-acceptance, no Sev-1 / Sev-2 bugs outstanding	30%

10.3 Technical stack

Vendor must pick from the following — no exotic frameworks without prior approval:

Layer	Allowed
Backend	PHP/Laravel, Node.js/Express, Python/Django
Frontend	Static HTML/CSS/JavaScript (the prototype's stack). No SPA framework.
Database	MySQL 8+, PostgreSQL 14+
Hosting	Linux VPS (DigitalOcean droplet, AWS Lightsail, equivalent). Sub-\$20/month tier acceptable for Phase 1 staging.
Build tools	None required; if used, must be free/open-source and not introduce licence costs
Proprietary CMS	Not permitted (no Magento, Adobe Experience Manager, Oracle, SAP) without [Customer] approval

10.4 Source-code ownership

All source code, design assets, database schemas, deployment scripts, API specifications, technical design documents, and account credentials produced under this engagement are work-for-hire and become **[Customer]'s exclusive property on final payment for the corresponding phase**. The vendor retains no licence, no right to reuse the code or designs in other engagements, and no right to host any artefact beyond the warranty period.

10.5 Warranty

30 days from Phase 1 acceptance. Vendor fixes any:

- **Sev-1** (application non-functional, data loss risk) — acknowledgment within 4 working hours, fix within 24 hours
- **Sev-2** (major flow broken, security defect) — acknowledgment within 1 working day, fix within 5 working days
- **Sev-3** (cosmetic / minor) — fix in next scheduled release window

...at no additional charge.

10.6 Project management

[Customer] expects:

- Single point of contact (project lead) at vendor — name + email + phone in the proposal
- Weekly status email (max half-page, plain text)
- Staging URL accessible to [Customer] from end of week 2 onwards
- No requirement for Gantt charts, Jira dashboards, or formal PM artefacts
- Change requests handled via written addendum (no informal scope changes)

10.7 Proposal expectations

The vendor's proposal should include:

1. Total fixed price for Phase 1 (BDT, exclusive of VAT)
2. Indicative duration (weeks from kickoff to acceptance demo)
3. Named team (project lead + ≥ 1 backend + ≥ 1 front-end developer)
4. Technical-design document outline (1–2 pages, not the full doc)
5. References — at least 2 prior BD-NBFI / fintech projects
6. Assumptions made by vendor (any deviations from §9.2)

Appendix A — Translation string sample

A small subset, for the vendor's reference. Full set in `translations.js` :

Key	English	বাংলা
login	Customer Login	গ্রাহক লগইন
apply	Apply for Loan	ঋণের আবেদন
brand_name	IDLC	আইডিএলসি
choose_loan_type	Choose your loan type	আপনার ঋণের ধরন নির্বাচন করুন
prod_home_loan	Home Loan	গৃহ ঋণ
prod_car_loan	Car Loan	গাড়ি ঋণ
prod_personal_loan	Personal Loan	ব্যক্তিগত ঋণ
prod_sme_loan	SME Loan	এসএমই ঋণ
modal_purpose_label	Purpose of the Loan	ঋণের উদ্দেশ্য
modal_financing_label	Financing Mode	অর্থায়ন পদ্ধতি
modal_financing_conventional	Conventional Financing	প্রচলিত অর্থায়ন
modal_financing_islamic	Islamic Financing	ইসলামিক অর্থায়ন
request_otp	Request OTP	ওটিপি অনুরোধ করুন
submit_otp	Submit OTP	ওটিপি জমা দিন
form_submit	Submit Application	আবেদন জমা দিন
app_submitted_title	Application Submitted	আবেদন জমা দেওয়া হয়েছে
app_ineligible_title	Application Could Not Be Processed	আবেদন প্রক্রিয়া করা যায়নি
dob_age_required	Applicants must be at least 18 years of age to submit this loan application.	এই ঋণের আবেদন জমা দেওয়ার জন্য আবেদনকারীর বয়স কমপক্ষে ১৮ বছর হতে হবে।

Full file: `translations.js` in the repository — ~250 keys, two languages.

Appendix B — Tracking-number format

IDLC-YYMMDD-XXXX

```
|
|   └─ 4-char alphanumeric, uppercase
|   |   Charset: A-Z + 2-9 (no 0, O, 1, I, L)
|   └─ Date of submission: YY MM DD
└─ Fixed brand prefix (replace at production-brand cutover)
```

Example: IDLC-260520-K7P4

Tracking numbers are:

- Issued only on `eligible` and `reduced` outcomes (never on `ineligible`)
- Globally unique (database UNIQUE constraint)
- Generated server-side (never client-side)
- Returned in the API response and rendered to the visitor on success
- Stored in `applications.tracking_number`

Appendix C — Glossary

Term	Meaning
NBFI	Non-Banking Financial Institution
CIF	Customer Information File — legacy term for a customer's master record
CIB	Credit Information Bureau — Bangladesh Bank's credit-reporting body
NID	National Identity Document — Bangladesh national ID
EMI	Equated Monthly Instalment
KYC	Know Your Customer
MFS	Mobile Financial Services (bKash, Nagad, Rocket, etc.)
BB	Bangladesh Bank — central bank
LTV	Loan to Value
DTI	Debt to Income
OTP	One-Time Password
BDT	Bangladeshi Taka (currency)
OWASP	Open Web Application Security Project
HSTS	HTTP Strict Transport Security
CSP	Content Security Policy
WCAG	Web Content Accessibility Guidelines
Sev-1 / 2 / 3	Severity classifications for defects
Phase 1 / 2 / 3	Engagement phases — MVP / Integration / Enhancements

Appendix D — Sample SQL DDL

```
CREATE TABLE applications (  
  id UUID PRIMARY KEY,  
  tracking_number VARCHAR(20) UNIQUE,  
  product VARCHAR(20) NOT NULL,  
  purpose VARCHAR(120),  
  financing_mode VARCHAR(20),  
  name VARCHAR(120) NOT NULL,  
  nid VARCHAR(20) NOT NULL,  
  dob DATE NOT NULL,  
  phone VARCHAR(15) NOT NULL,  
  cif VARCHAR(20),  
  email VARCHAR(120),  
  profession VARCHAR(20),  
  monthly_income DECIMAL(15,2),  
  has_burden BOOLEAN,  
  monthly_burden DECIMAL(15,2),  
  expected_amount DECIMAL(15,2),  
  loan_term_years INT,  
  eligibility_status VARCHAR(15),  
  eligible_amount DECIMAL(15,2),  
  language VARCHAR(2),  
  declarations_accepted_at TIMESTAMP,  
  ip_address VARCHAR(45),  
  user_agent TEXT,  
  created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,  
  updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP  
);  
  
CREATE INDEX idx_applications_nid ON applications(nid);  
CREATE INDEX idx_applications_phone ON applications(phone);  
CREATE UNIQUE INDEX idx_applications_tracking ON applications(tracking_number);  
  
CREATE TABLE audit_log (  
  id UUID PRIMARY KEY,  
  application_id UUID REFERENCES applications(id),  
  event_type VARCHAR(40),  
  actor VARCHAR(120),  
  timestamp TIMESTAMP DEFAULT CURRENT_TIMESTAMP,  
  payload JSON  
);  
  
CREATE TABLE otp_requests (  
  id UUID PRIMARY KEY,  
  phone VARCHAR(15),  
  code_hash VARCHAR(64),  
  requested_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,  
  expires_at TIMESTAMP,  
  verified_at TIMESTAMP,  
  attempt_count INT DEFAULT 0
```

```
);  
  
CREATE TABLE admin_users (  
  id UUID PRIMARY KEY,  
  email VARCHAR(120) UNIQUE,  
  password_hash VARCHAR(120),  
  role VARCHAR(20),  
  created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,  
  last_login_at TIMESTAMP  
);
```

End of document.